

11. **Assertion (A):** Metals have lustre.

Reason (R): Metals have loosely bound electrons.

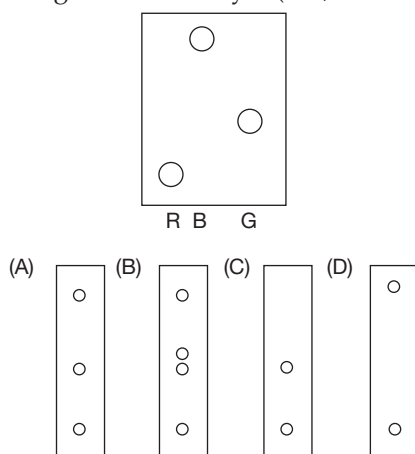
- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

12. **Assertion (A):** Steel is more resistant to corrosion.

Reason (R): Alloying makes the constituent metal less reactive.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

13. A standard chromatogram of three dyes (red, blue and green) is given below.



Based on the standard chromatogram analysis and the given chromatograms (A, B, C and D), arrange them in the order of the descriptions given below.

- (i) This chromatogram represents a mixture of all the three dyes along with an impurity.
 - (ii) This chromatogram represents a mixture of red and blue dyes.
 - (iii) This chromatogram represents a mixture of all the three dyes.
 - (iv) This chromatogram represents a mixture of red and green dyes.
- (a) CABD (b) BDAC (c) ADCB (d) DBAC

14. A saturated solution of NaCl and KNO_3 is taken in a beaker. Some amount of carbon tetrachloride is added to it. Then sand is added to it. Four processes given below are carried out to separate the constituents of the above mixture. Arrange them in the proper sequence.

2.4 Chapter 2 Classification of Matter

- (A) Evaporation (B) Fractional crystallization
(C) Separating funnel (D) Filtration
(a) ACBD (b) BDAC (c) CABD (d) DCBA

15. Match the statements of Column A with those of Column B.

Column A (Mixture)	Column B (Category)
(i) Copper sulphate crystals are dropped in water and stirred.	(A) Liquid–liquid heterogeneous
(ii) Some amount of benzene is poured into water and stirred well.	(B) Liquid–liquid homogeneous
(iii) Finely powdered sulphur is added to water and stirred well.	(C) Solid–liquid homogeneous
(iv) Some amount of vinegar is poured into water and stirred.	(D) Solid–solid homogeneous
(v) Iron is alloyed to manufacture steel.	(E) Solid–liquid heterogeneous

- (a) (i) → (E); (ii) → (D); (iii) → (A); (iv) → (B); (v) → (C)
(b) (i) → (A); (ii) → (C); (iii) → (B); (iv) → (E); (v) → (D)
(c) (i) → (C); (ii) → (A); (iii) → (E); (iv) → (B); (v) → (D)
(d) (i) → (B); (ii) → (E); (iii) → (A); (iv) → (C); (v) → (D)

Space for rough work

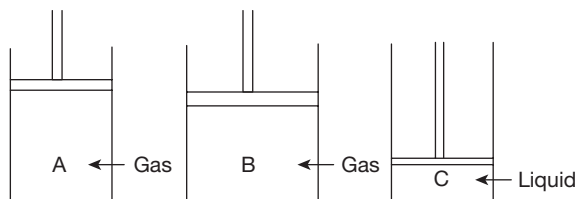
Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

- Ice is kept in a vacuum chamber. In which state will it exist?
(a) Solid (b) Liquid (c) Vapour (d) Gas
- Which of the following interconversions of states of matter is involved in filling up of LPG cylinder?
(a) Sublimation (b) Liquefaction
(c) Condensation (d) Evaporation
- An element is a good conductor of electricity, lustrous, and it is used as an electrode in electrometallurgical processes due to its high resistance to heat and chemicals. The constituent atom of the element is _____.
(a) Cu (b) Fe (c) C (d) Si
- Which of the following is a homogeneous mixture?
(a) Brass (b) Milk
(c) Gun powder (d) Muddy water
- Find the odd one out with respect to the conduction of electricity.
(a) Red phosphorus (b) Rhombic sulphur
(c) Gas carbon (d) Diamond
- Finely powdered charcoal is added to coconut oil. This mixture is added to glucose solution. How will you separate them?
(a) Filtration, separating funnel and evaporation
(b) Evaporation, separating funnel and filtration
(c) Filtration, followed by evaporation
(d) Separating funnel, followed by filtration
- We blow air to cool a hot liquid in order to _____.
(a) create turbulence in the hot liquid
(b) provide a larger surface area for the liquid
(c) provide fresh air above the surface of the liquid
(d) saturate the air above the liquid with water vapour
- The same amount of three gases 'A', 'B' and 'C' is taken in three different cylinders. All the cylinders are fitted with movable pistons and the gases are kept at room temperature and normal pressure. Then, external pressure is applied on each piston. The changes in the volumes are shown in the figures given below.



Which of the following statements is correct regarding the above processes?

- (a) A has high critical temperature than B.
- (b) Critical temperature of B is above the room temperature.
- (c) Critical temperature of C is the highest.
- (d) Intermolecular forces of attraction among the molecules of gas A is the maximum.

9. In which of the following, does kinetic energy decrease?

- (a) Steam at 100°C gets converted to superheated steam.
- (b) A cup of hot tea at 60°C gets cold.
- (c) Steam at 100°C getting condensed in distillation process.
- (d) Water is getting converted to ice at 0°C in a refrigerator.

10. In which of the following cases, change of state of matter takes place due to the application of pressure?

- (a) Skating on ice
- (b) Formation of snow
- (c) Filling up of cooking gas (LPG) cylinder
- (d) Both (a) and (c)

11. **Assertion (A):** Metals are good conductors of electricity.

Reason (R): Metal atoms have completely filled valence shells.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

12. **Assertion (A):** Duralumin is as strong as steel.

Reason (R): A metal becomes more malleable when it is converted to an alloy.

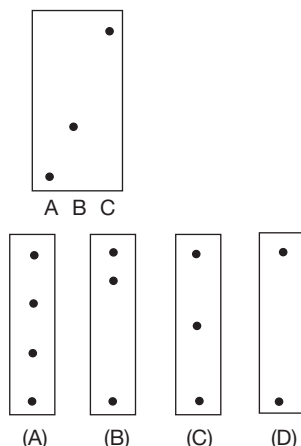
- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

13. A standard chromatogram of the constituents of a food colour is given below.

Four chromatograms (A, B, C and D) shown below represent four different brands of food colours available in the market.

Arrange them in the order of the descriptions given below.

Space for rough work



- (i) This food colour is constituted of A and C along with another component.
 (ii) This food colour is constituted of all three components—A, B and C.
 (iii) The food colour is constituted of A and B, but it contains two other components as well.
 (iv) This food colour is constituted of A and C only.
- (a) ACBD (b) CDBA (c) BCAD (d) DBAC

14. A solution of common salt is taken in a beaker. Some amount of vinegar is poured into it. Then, a mixture of sawdust and coarsely ground chalk is added to it. Four processes given below are carried out to separate the constituents of the above mixture. Arrange the processes in proper sequence in order to separate the components one by one.

- (a) Fractional distillation (b) Filtration
 (c) Evaporation (d) Gravity separation
- (a) DBAC (b) BDAC (c) CABD (d) ABCD

15. Match the statements of Column A with those of Column B.

Column A (Types of Mixtures)	Column B (Examples)
(i) Solid–solid homogeneous mixture	(A) Coconut oil and water
(ii) Solid–liquid homogeneous mixture	(B) Petrol and diesel
(iii) Liquid–liquid heterogeneous mixture	(C) Bronze
(iv) Liquid–liquid homogeneous mixture	(D) Finely powdered charcoal in water
(v) Solid–liquid heterogeneous mixture	(E) Saline water

- (a) (i) → (C); (ii) → (E); (iii) → (A); (iv) → (B); (v) → (D)
 (b) (i) → (B); (ii) → (C); (iii) → (A); (iv) → (D); (v) → (E)
 (c) (i) → (E); (ii) → (A); (iii) → (B); (iv) → (C); (v) → (D)
 (d) (i) → (D); (ii) → (B); (iii) → (A); (iv) → (E); (v) → (C)

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Which of the following statements is true regarding thermal expansion?
(a) Solids show maximum thermal expansion.
(b) Liquids show maximum thermal expansion.
(c) Gases show maximum thermal expansion.
(d) Thermal expansion does not depend on the state of a substance.
2. Which of the following does not help to distinguish evaporation and boiling?
(a) Temperature at which the phenomenon occurs
(b) Portion of the liquid from which they occur
(c) Change of state of matter
(d) Speed at which they occur
3. Good conductance of electricity is the characteristic of a _____.
(a) metal (b) non-metal
(c) metalloid (d) solid substance
4. Arrange the following in increasing order of atomicity.
(A) Phosphorus (B) Iodine
(C) Sulphur (D) Ozone
(a) BDAC (b) DBCA (c) DBAC (d) ACBD
5. Arrange the following alloys in the order of their major components given below.
(copper, iron, nickel and aluminium)
(A) Monel metal (A) Duralumin
(C) German silver (D) Tungsten steel
(a) DCAB (b) CDAB (c) BCDA (d) CDBA
6. Which of the following processes can separate sodium chloride and potassium nitrate?
(a) Sublimation (b) Fractional distillation
(c) Distillation (d) Fractional crystallization
7. Which of the following has the least density?
(a) Potassium (b) Water
(c) Mercury (d) Copper

Space for rough work

8. **Assertion (A):** A mixture of alcohol and water can be separated by using a separating funnel.

Reason (R): Alcohol and water are miscible liquids.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

9. **Assertion (A):** Critical temperature of CO_2 is greater than that of O_2 .

Reason (R): Intermolecular force of attraction of CO_2 is more than that of O_2 .

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

10. Match the entries of Column A with those of Column B.

Column A (Mixture)	Column B (Principle of Suitable Separation Process)
(i) $\text{NH}_4\text{Cl} + \text{KNO}_3$	(A) Difference in the boiling points of components in their liquid state.
(ii) Sulphur + carbon disulphide	(B) Decrease in solubility of a gas in liquid with the decrease in pressure.
(iii) Soda water	(C) One of the components can sublime on heating
(iv) Air	(D) Non-sublimable solid solute in liquid solvent.

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
 (b) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
 (c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (C); (iv) $\rightarrow$ (A)
 (d) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)

11. The order of critical temperature of four gases A, B, C and D is as follows:

$$A > D > C > B$$

Which of the following has the least vapour pressure?

- (a) A (b) B (c) C (d) D

12. The vapour pressure of liquid A is higher than that of liquid B at a particular temperature. Which of the following statements is true?

- (a) The boiling point of A is greater than that of B.
 (b) The boiling point of B is greater than that of A.
 (c) They have the same boiling point.
 (d) The information provided is not sufficient to compare their boiling points.

13. Which of the following is a heterogeneous mixture?
- (a) Milk (b) Soda water
(c) Solution of sugar (d) Mixture of O_2 and H_2
14. The boiling point of which of the following is the maximum under normal atmospheric pressure?
- (a) Distilled water
(b) Tap water
(c) Sea water
(d) Rain water
15. Skating on ice is possible because _____.
- (a) increase in pressure increases the melting point of ice
(b) increase in pressure decreases the melting point of ice
(c) the volume of ice expands when it melts
(d) the density of ice decreases on the application of pressure

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. If the same amount of heat is supplied to the following substances, which shows the least expansion?
 - (a) Iron block
 - (b) Water
 - (c) Oil
 - (d) O_2 gas kept in a cylinder with a movable piston
2. Which of the following is not a characteristic of boiling?
 - (a) It is a bulk phenomenon.
 - (b) It occurs at a particular temperature.
 - (c) Temperature remains constant during the process.
 - (d) It is a very slow process.
3. Which of the following liquids can conduct electricity in pure state?
 - (a) Water
 - (b) Mercury
 - (c) Bromine
 - (d) Benzene
4. Which of the following is monoatomic?
 - (a) Hydrogen
 - (b) Chlorine
 - (c) Helium
 - (d) Nitrogen
5. Which of the following alloys contains copper as the major component?
 - (a) Stainless steel
 - (b) Monel metal
 - (c) Bell metal
 - (d) Magnalium
6. Arrange the following processes of separation in the order of the mixtures given below:
sand + sugar; components of black ink; iron + iodine; powder of naphthalene + common salt
 - (A) Magnetic separation
 - (B) Sublimation
 - (C) Chromatography
 - (D) Solvent extraction
 - (a) CDBA
 - (b) ABCD
 - (c) DCAB
 - (d) BCDA
7. Arrange the following in the ascending order of their densities.
 - (A) Water
 - (B) O_2 gas
 - (C) A solid block of aluminium
 - (D) Ice
 - (a) ADCB
 - (b) BDCA
 - (c) ACBD
 - (d) BDAC

8. **Assertion (A):** Ammonium chloride cannot be separated from water by distillation.

Reason (R): NH_4Cl dissolves in water.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

9. **Assertion (A):** NH_3 is an easily liquefiable gas.

Reason (R): The intermolecular forces of attraction are very less in NH_3 .

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

10. Match the statements of Column A with those of Column B.

Column A (Process of Separation)	Column B (Principle Involved in the Process of Separation)
(i) Distillation	(A) Difference in the ease of liquefaction of the component gases.
(ii) Filtration	(B) Immiscible liquids present in distinct layers.
(iii) Preferential liquefaction	(C) Insoluble solid component present in a liquid.
(iv) Separating funnel	(D) Non-sublimable solid solute in liquid solvent and both can be obtained after separation.

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
- (b) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
- (c) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
- (d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)

11. Gas A can be liquefied easily, but liquefaction of gas B is difficult at a particular temperature. Which of the following conclusions can be drawn from the given statement?

- (a) Vapour pressure of gas A is more than that of gas B at a particular temperature.
- (b) Boiling point of A is lower than that of B.
- (c) Intermolecular force of attraction of A is more than that of B.
- (d) All the above